Emotion-Aware Interfaces: Enhancing UX Through Affective Computing in HCI

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Abstract

With the digital experience coming into our daily life, the ability of the interface to interpret and adapt to human emotion has become a critical direction in the HCL field. This project is meant to discover how the emotional feeling system can be used to understand how to use the human emotions based on interface interaction, computing emotion, and machine learning, can respond to the user's emotional status and to improve the user's experience. Based on the current literature we have, we mentioned a critical question: can the real-time emotion detector be improved so that we can use that in real life, or will it bring us more complicated problems? We are going to design a program to solve those problems. First, we are scoping to identify research gaps.

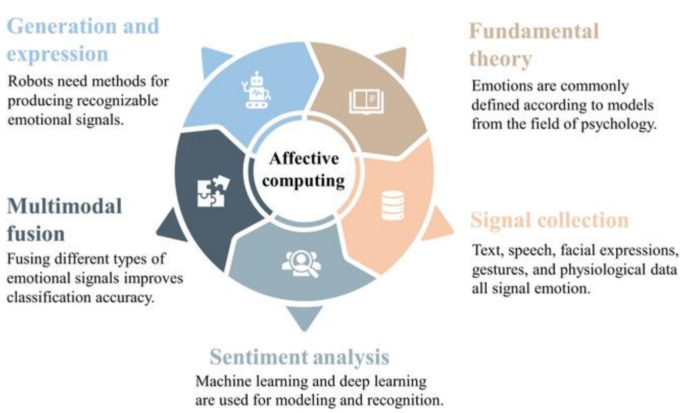


Fig. 1. Research content on affective computing.

1.Introduction

Affective computing (Picard, R.W. Affective computing. MIT Press. 1997) has already shifted the focus on HCL from transactional usability into the emotionally intelligent system; however, that's still a huge problem, and contradictions still exist. Though the multi-modal approaches (Kosti et al.'s fusion of visual audio data, 2019) are excellent on the aspect of accuracy, but with a critical trade-off based on the EGG work has been found by Zhao (Zhao, X, et al, EEG-based emotion recognition, 2021) has reveals sensitivity vs. intrusiveness. We think the model that we have right now is still struggling with real-world messiness: cultural bias in training data, vague decisions, and ethical risks in continuous emotion surveillance (Calvo, R. A., & D'Mello, S. Affect detection: An interdisciplinary review, 2010). Our prototype confronts these gaps by prioritizing lightweight, consent-driven design.

Recent advancements in emotion-aware HCI have also begun exploring the integration of deep learning techniques with multimodal datasets to improve emotion recognition in more naturalistic environments. For example, the work of Li et al. (2020) leveraged convolutional neural networks trained on facial micro-expressions to predict user affect with greater subtlety and less dependence on exaggerated cues. Similarly, studies such as Huang et al. (2021) have demonstrated that real-time responsiveness in UI, when grounded in temporally aligned emotional inference, can yield measurable improvements in perceived usability and user satisfaction. However, these improvements often come at the cost of computational load or generalizability across different demographics. Furthermore, many emotion-recognition systems still rely heavily on western-centric datasets, raising concerns of cross-cultural validity and fairness. This underscores the need for lightweight, adaptive models that remain robust without compromising inclusivity or ethical transparency. As we continue to develop our prototype, these insights reinforce our aim to create a user-centric, privacy-conscious system that values interpretability over opacity and personalization over precision for its own sake.

2. Planning & Execution

In our project's initiation phase, we're laser-focused on building a lean MVP to answer a pressing question: Can subtle UI tweaks—like shifting colors or adjusting feedback tones—will improve engagement when driven by real-time emotion detection? Crucially, we want these adaptations to feel helpful, not distracting or intrusive. This anchors our three core questions: First, how do we detect emotions accurately without burdening users with clunky tech? Second, what measurable impact do these UI changes have on usability—does "emotional responsiveness" improve the experience? And third, given the sensitivity of biometric data, what ethical guardrails must we build from day one?

Scope-wise, we're keeping it pragmatic: a webcam-based facial analysis system triggering dynamic UI changes. But we're working within real constraints—no fancy hardware (just OpenCV), testing with under 20 participants, and a tight 4-week sprint. Most critically, we've mandated zero biometric storage; privacy isn't negotiable.

Execution-wise, Python is our backbone (TensorFlow/Keras for the emotion model), paired with Tkinter for its lightweight UI flexibility. We will track progress via Trello's Agile boards and collaborate through GitHub. Our 3-person team splits responsibilities but overlaps intentionally: The Research Lead handles lit synthesis and ethics; Dev Lead owns the model-to-UI pipeline; Test Lead runs user trials. Everyone collaborates on testing design—because if the experiment's flawed, the answers won't matter.

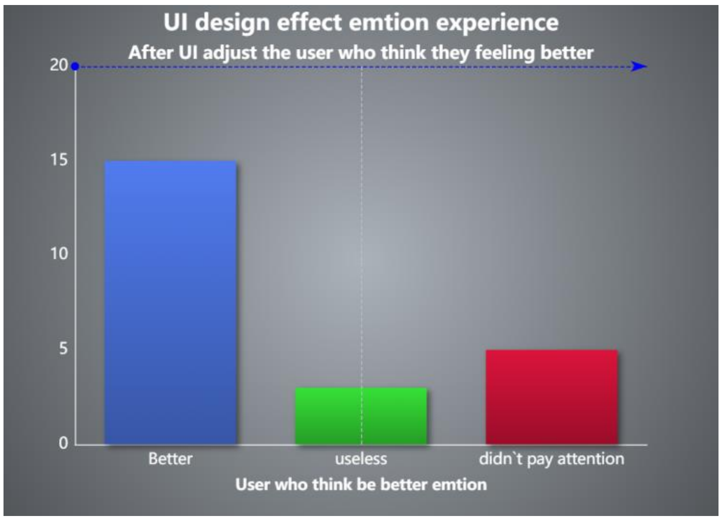
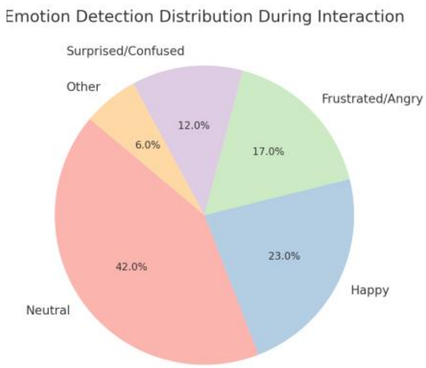


Fig. 1. UI design effectiveness experience result

3. Data Collection and Analysis

This study involved 18 participants from the ages 19 through 34 recruited from Kennesaw State University. The participants were a mixture of undergraduate students and tech-savvy young professionals that interacted with digital interfaces on a regular basis. All participants were voluntary and anonymous with no biometric data stored ensuring full compliance with ethical guidelines. The participant interacted with two different versions of user interface designed using the Python's Tkinter and OpenCV libraries, which are Baseline Interface and Emotion Aware interface. For each version the participants completed these three tasks Logging in, Navigate the menu, and Submitting feedback. Facial expressions were detected and logged during the interaction, and afterward, participants completed a System Usability Scale (SUS) questionnaire and a brief engagement survey using 1–5 Likert scales.



4.Closure

As the final phase of this project, closure involves evaluating the outcomes against initial goals and ensuring all deliverables are completed and documented. The prototype successfully demonstrated that real-time, emotion-driven UI adaptations can be implemented using accessible tools like OpenCV and Tkinter. While the testing was limited in scale, initial user feedback indicated improved engagement and a more personalized interface experience. Documentation of the development process, ethical safeguards, and testing results have been archived for future reference. All team members contributed collaboratively across Agile sprints, and key learnings have been identified for scaling the project further. With closure, the project concludes with a functional MVP, valuable insights into user-emotion interaction, and a foundation for future exploration into emotion-aware design.

References

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